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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a substrate, a source trench structure, a gate trench structure and a drain electrode. The source trench structure is in the substrate. The gate trench structure is in the substrate, and a bottom of the gate trench structure is higher than a bottom of the source trench structure. The gate trench structure surrounds the source trench structure and defines a device unit, and a first projection area of the source trench structure along a fixed direction is less than 20% of a second projection area of the device unit along the fixed direction. The drain electrode is below the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Taiwan Application Serial Number 113106518 filed Feb. 23, 2024, which is herein incorporated by reference.

BACKGROUND

Technical Field

[0002] Some embodiments of the present disclosure relate to a semiconductor device.

Description of Related Art

[0003] Metal oxide semiconductor field effect transistors (MOSFETs) can be classified into horizontal-channel MOSFETs and vertical-channel MOSFETs according to the channel direction thereof. The vertical-channel MOSFETs can provide the same current with a relatively small area while obtaining a relatively low on-resistance, thereby remarkably reducing the production cost. Thus, further improving the efficacy of vertical-channel MOSFETs has also become one of important issues.

SUMMARY

[0004] Some embodiments of the present disclosure provide a semiconductor device, including a substrate, a source trench structure, a gate trench structure and a drain electrode. The source trench structure is in the substrate. The gate trench structure is in the substrate, a bottom of the gate trench structure being higher than a bottom of the source trench structure, where the gate trench structure surrounds the source trench structure and defines a device unit, and a first projection area of the source trench structure along a fixed direction is less than 20% of a second projection area of the device unit along the fixed direction. The drain electrode is below the substrate.

[0005] In some embodiments, the first projection area of the source trench structure is less than 15% of the second projection area of the device unit.

[0006] In some embodiments, the first projection area of the source trench structure is more than 5% of the second projection area of the device unit.

[0007] In some embodiments, the device unit has a shape of hexagon, quadrangle or triangle in a top view.

[0008] In some embodiments, the device unit has a shape of hexagon in a top view, and the hexagon has a first side and a second side, where the first side has a greater length than the second side.

[0009] In some embodiments, four sides of the hexagon have a first length, and the remaining two sides have a second length different from the first length.

[0010] In some embodiments, the substrate further includes a source doped region, between the gate trench structure and the source trench structure.

[0011] In some embodiments, a source contact extends from an upper surface of the source doped region to an upper surface of the source trench structure.

[0012] In some embodiments, the source trench structure includes a dielectric layer and a conductor layer, and the conductor layer is surrounded by the dielectric layer.

[0013] In some embodiments, the gate trench structure includes a gate dielectric layer and a gate layer, and the gate layer is surrounded by the gate dielectric layer.

[0014] Some embodiments of the present disclosure provide a semiconductor device, including a substrate, a source trench structure, a gate trench structure and a drain electrode. The substrate includes a source region. The source trench structure is in the substrate. The gate trench structure is in the substrate and adjacent to the source region, a bottom of the gate trench structure being higher than a bottom of the source trench structure, where the gate trench structure surrounds the source trench structure and defines a device unit, and an interface between the gate trench structure and

the source region of the substrate along a fixed direction is a hexagon. The drain electrode is below the substrate.
[0015] In some embodiments, the source region is between the gate trench structure and the source trench structure.
[0016] In some embodiments, the six sides of the hexagon have substantively the same length.
[0017] In some embodiments, the hexagon has a first side and a second side, where the first side has a greater length than the second side.
[0018] In some embodiments, the first projection area of the source trench structure along a fixed direction is 5% to 20% of the second projection area of the device unit along the fixed direction.
[0019] In some embodiments, the first projection area of the source trench structure is 5% to 15% of the second projection area of the device unit along a fixed direction.
[0020] In some embodiments, four sides of the hexagon have a first length, and the remaining two sides have a second length different from the first length.
[0021] In some embodiments, the source region is a source doped region, between the gate trench structure and the source trench structure.
[0022] In some embodiments, the source trench structure includes a dielectric layer and a conductor layer, and the conductor layer is surrounded by the dielectric layer.
[0023] In some embodiments, the gate trench structure includes a gate dielectric layer and a gate layer, and the gate layer is surrounded by the gate dielectric layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] FIG. **1** is a top view of a semiconductor device according to some embodiments of the present disclosure.
[0025] FIG. **2** is a cross-sectional view along a line A-A′ of FIG. **1**.
[0026] FIG. **3** is a top view of a semiconductor device according to some other embodiments of the present disclosure.
[0027] FIG. **4** is a top view of a semiconductor device according to some other embodiments of the present disclosure.
[0028] FIG. **5** is a top view of a semiconductor device according to some other embodiments of the present disclosure.
DETAILED DESCRIPTION
[0029] Some embodiments of the present disclosure relate to a double-trench semiconductor device, for example, a double-trench metal-oxide-semiconductor field-effect transistor (MOSFET). The double-trench semiconductor device may include a source trench structure and a gate trench structure, and the source trench structure can be used to protect the corner of the gate trench structure to avoid problems concerning the reliability of the gate trench structure. In the present disclosure, the top view area of the source trench structure can be controlled at a proper size so that the on-resistance of the semiconductor device can be maintained low while the source trench structure still has ability of protecting the corner of the gate trench structure.
[0030] FIG. **1** is a top view of a semiconductor device **100** according to some embodiments of the present disclosure. Referring to FIG. **1**, the semiconductor device **100** includes a plurality of device units U, and the device units U can be closely arranged together. For instance, all boundaries of one of the device units U contact the boundaries of other device units U. In some embodiments, in the top view, the device unit U has a shape of hexagon in the top view, as shown in FIG. **1**. It should be noted that, for simplicity of illustration, FIG. **1** only depicts some of the components.
[0031] FIG. **2** is a cross-sectional view along a line A-A′ of FIG. **1**. Referring to FIG.
[0032] **2**, the semiconductor device **100** may include a substrate **110**, a source trench structure **120**,

a source contact **130**, a gate trench structure **140** and a drain electrode **150**.

[0033] The substrate **110** includes a drift region **112**, a body contact region **114**, a well region **116** and a source region **118** (also referred to as a source doped region **118**). The body contact region **114**, the well region **116** and the source region **118** are on the drift region **112**. The body contact region **114** extends downward from an upper surface of the substrate **110**. The well region **116** transversely surrounds the body contact region **114**. The source region **118** transversely surrounds the body contact region **114** and is on the well region **116**. In some embodiments, the drift region **112** and the source region **118** have a first conductivity type, the well region **116** and the body contact region **114** have a second conductivity type, and the first conductivity type is different from the second conductivity type. In some embodiments, the source region **118** is an N-type heavily-doped region, the drift region **112** is an N-type lightly-doped region, the well region **116** is a P-type lightly-doped region or moderately-doped region, and the body contact region **114** is a P-type heavily-doped region. In some embodiments, the substrate **110** is made from silicon or silicon carbide.

[0034] The source trench structure **120** is in the substrate **110**, and is surrounded by the body contact region **114**. In some embodiments, an interface between the source trench structure **120** and the body contact region **114** is a circular outline. The source trench structure **120** includes a dielectric layer **122** and a semiconductor layer (or a conductor layer) **124**. The semiconductor layer **124** is surrounded by the dielectric layer **122**, and the dielectric layer **122** is surrounded by the body contact region **114**. In other words, the dielectric layer **122** is between the body contact region **114** and the semiconductor layer **124**. In some embodiments, the dielectric layer **122** can be made from silicon oxide, silicon nitride or other analogues. In some embodiments, the semiconductor layer **124** is made from polycrystalline silicon. The source contact **130** is on the source trench structure **120**, and further extends from an upper surface of the source region **118** to an upper surface of the source trench structure **120**. In some embodiments, the source contact **130** can be made from a metal.

[0035] The gate trench structure **140** is in the substrate **110**, and surrounds the source trench structure **120**, the well region **116** and the source region **118**. Therefore, the source region **118** is between the source trench structure **120** and the gate trench structure **140**. The gate trench structure **140** includes a gate dielectric layer **142** and a gate layer **144**. The gate layer **144** is surrounded by the gate dielectric layer **142**, and the gate dielectric layer **142** contacts the well region **116** and the source region **118**. In other words, the well region **116** and the source region **118** are between the source trench structure **120** and the gate trench structure **140**. A bottom of the gate trench structure **140** is higher than a bottom of the source trench structure **120**. In some embodiments, the gate dielectric layer **142** can be made from silicon oxide, silicon nitride or other analogues. In some embodiments, the gate layer **144** is made from a semiconductor or conductor, for example, polycrystalline silicon. In some embodiments, an interface between the well region **116** and the gate trench structure **140** is a hexagon. In some embodiments, an interface between the source region **118** and the gate trench structure **140** has a different shape from an interface between the source trench structure **120** and the body contact region **114**.

[0036] The drain electrode **150** is below the substrate **110**. In some embodiments, the substrate **110** can be otherwise formed on a heavily-doped substrate having a first conductivity type. Therefore, the drain electrode **150** and the substrate **110** can be formed at two opposite sides of the heavily-doped substrate having the first conductivity type. In some embodiments, the drain electrode **150** can be made from a metal.

[0037] The semiconductor device **100** of the present disclosure is a vertical-channel device, which can increase the current on a unit area. Specifically, the semiconductor device of the present disclosure consists of a plurality of device units U, and in the top view (FIG. **1**, for instance), the gate trench structure **140** defines a device unit U. It should be noted that, although there is a distinct boundary between the device units U in FIG. **1**, it is merely intended for clearly showing the range of each device unit U. In fact, there is no distinct boundary between the device units U. For example, adjacent device units U can share a gate trench structure **140**. Thus, the top view area of each of the device units U can be defined as an area enclosed by a line connecting the midpoints of the gate trench structure **140**. When the semiconductor device **100** is in operation, in each of the device units U, an electron flow can flow downward to the drift region **112** from the source contact **130** along the source region **118** and the well region **116** on a side wall of the gate trench structure **140**, and then flow to the drain electrode **150**. Now, a high electric field of the drain electrode **150** may induce a problem concerning reliability of the gate dielectric layer **142** at the corner of the gate trench structure **140**.

[0038] The source trench structure **120** and the body contact region **114** around the source trench structure **120** can be used to protect the corner region of the gate trench structure **140**. Specifically, when the semiconductor device **100** is in operation, the source trench structure **120** is electrically connected to the source contact **130**, the body contact region **114** around the source trench structure **120** has a second conductivity type while the drift region **112** has a first conductivity type, and a depletion region is formed between the body contact region **114** and the drift region **112** to protect the corner region of the gate trench structure **140**. Therefore, the impact of the high electric field of the drain electrode **150** on the corner region of the gate trench structure **140** can be diminished. Since the source trench structure **120** per se cannot provide current, controlling the dimensions of the source trench structure **120** within an appropriate range can properly protect the corner region of the gate trench structure **140** while the dimensions of the semiconductor device **100** are reduced. In some embodiments, the first projection area of the source trench structure **120** along a fixed direction is less than 20% of the second projection area of the device unit U along the fixed direction (FIG. **1**). In some embodiments, the first projection area of the source trench structure **120** is more than 5% of the second projection area of the device unit U (FIG. **1**). When the proportion of the first projection area of the source trench structure **120** is beyond the above-mentioned range, the source trench structure **120** may occupy a larger part, making it impossible to reduce the dimensions of the semiconductor device **100** and hence the on-resistance of the semiconductor device **100**. When the proportion of the first projection area of the source trench structure **120** is below the above-mentioned range, the source trench structure **120** may be too far from the gate trench structure **140**, and thus the depletion region formed by the body contact region **114** and the drift region **112** around the source trench structure **120** cannot effectively protect the corner region of the gate trench structure **140**. In some embodiments, the first projection area of the source trench structure **120** along a fixed direction and the second projection area of the device unit U along the fixed direction are shown in FIG. **1**.

[0039] FIG. **3** is a top view of a semiconductor device **100** according to some other embodiments of the present disclosure. A cross-sectional view along a line A-A′ of FIG. **3** can also be shown in FIG. **2**. The semiconductor device **100** of FIG. **3** is similar to the semiconductor device **100** of FIG. **1**, with a difference in that the six sides of the hexagon of the device unit U of the semiconductor device **100** of FIG. **1** have substantively the same length. On the other hand, the device unit U of the semiconductor device **100** of FIG. **3** has a first side S**1**, a second side S**2**, a third side S**3**, a fourth side S**4**, a fifth side S**5** and a sixth side S**6** connected clockwise, where four sides of the hexagon (for example, the second side S**2**, the third side S**3**, the fifth side S**5** and the sixth side S**6**) have substantively the same first length, and the remaining two sides (for example, the first side S**1** and the fourth side S**4**) of the hexagon have a second length different from the first length. In some embodiments, the length of the first side S**1** and the fourth side S**4** is greater than the length of the second side S**2**, the third side S**3**, the fifth side S**5** and the sixth side S**6**. When the top view of the semiconductor device **100** is shown in FIG. **3**, the proportion of the top view area of the source trench structure **120** based on the top view area of the device unit U can be further reduced relative to FIG. **1**. For example, the first projection area of the source trench structure **120** is less than 15% of the second projection area of the device unit U, but such proportion can still be more than 5%.

Therefore, the source trench structure **120** can properly protect the corner region of the gate trench structure **140** while the dimensions of the semiconductor device **100** are reduced.

[0040] FIG. **4** is a top view of a semiconductor device **100** according to some other embodiments of the present disclosure. A cross-sectional view along a line A-A′ of FIG. **4** can also be shown in FIG. **2**. The semiconductor device **100** of FIG. **4** is similar to the semiconductor device **100** of FIG. **1**, with a difference in that the device unit U of the semiconductor device **100** of FIG. **4** has a shape of rhombus in the top view. In some embodiments, an interface between the source region **118** and the gate trench structure **140** is also a rhombus. When the device unit U has a shape of rhombus in the top view, the on-resistance of the semiconductor device will be further reduced.

[0041] FIG. **5** is a top view of a semiconductor device **100** according to some other embodiments of the present disclosure. A cross-sectional view along a line A-A′ of FIG. **5** can also be shown in FIG. **2**. The semiconductor device **100** of FIG. **5** is similar to the semiconductor device **100** of FIG. **1**, with a difference in that the device unit U of the semiconductor device **100** of FIG. **5** has a shape of triangle in the top view. In some embodiments, an interface between the source region **118** and the gate trench structure **140** is also a triangle. When the device unit U has a shape of triangle in the top view, the on resistance of the semiconductor device will be further reduced.

[0042] It should be noted that, although some embodiments of the present disclosure provide device units U having different top-view shapes, the present disclosure is not limited thereto. As long as the proportion of the first protection area of the source trench structure **120** in the semiconductor device **100** based on the second projection area of the device unit U is within the scope of protection of the present disclosure, it can fall within the scope of protection of the present disclosure. In some embodiments of the present disclosure, the first projection area of the source trench structure **120** is less than 20% and more than 5% of the second projection area of the device unit U. Therefore, the source trench structure **120** can properly protect the corner region of the gate trench structure **140** while the dimensions of the semiconductor device **100** are reduced.

Claims

1. A semiconductor device, comprising: a substrate; a source trench structure, in the substrate; a gate trench structure, in the substrate, a bottom of the gate trench structure being higher than a bottom of the source trench structure, wherein the gate trench structure surrounds the source trench structure and defines a device unit, wherein a first projection area of the source trench structure along a fixed direction is less than 20% of a second projection area of the gate trench structure along the fixed direction; and a drain electrode, below the substrate.

2. The semiconductor device of claim 1, wherein the first projection area of the source trench structure is less than 15% of the second projection area of the device unit.

3. The semiconductor device of claim 1, wherein the first projection area of the source trench structure is more than 5% of the second projection area of the device unit.

4. The semiconductor device of claim 1, wherein the device unit has a shape of hexagon, quadrangle or triangle in a top view.

5. The semiconductor device of claim 1, wherein the device unit has a shape of hexagon in a top view, and the hexagon has a first side and a second side, wherein the first side has a greater length than the second side.

6. The semiconductor device of claim 5, wherein four sides of the hexagon have a first length, and the remaining two sides have a second length different from the first length.

7. The semiconductor device of claim 1, wherein the substrate further comprises: a source doped region, between the gate trench structure and the source trench structure.

8. The semiconductor device of claim 7, further comprising: a source contact extending from an upper surface of the source doped region to an upper surface of the source trench structure.

9. The semiconductor device of claim 1, wherein the source trench structure comprises: a dielectric layer; and a conductor layer, surrounded by the dielectric layer.

10. The semiconductor device of claim 1, wherein the gate trench structure comprises: a gate dielectric layer; and a gate layer, surrounded by the gate dielectric layer.

11. A semiconductor device, comprising: a substrate, comprising a source region; a source trench structure, in the substrate; a gate trench structure, in the substrate and adjacent to the source region, a bottom of the gate trench structure being higher than a bottom of the source trench structure, wherein the gate trench structure surrounds the source trench structure and defines a device unit, and an interface between the gate trench structure and the source region of the substrate along a fixed direction is a hexagon; and a drain electrode, below the substrate.

12. The semiconductor device of claim 11, wherein the source region is between the gate trench structure and the source trench structure.

13. The semiconductor device of claim 12, further comprising: a source contact extending from an upper surface of the source region to an upper surface of the source trench structure.

14. The semiconductor device of claim 11, wherein six sides of the hexagon have substantively the same length.

15. The semiconductor device of claim 11, wherein the hexagon has a first side and a second side, and the first side has a greater length than the second side.

16. The semiconductor device of claim 11, wherein a first projection area of the source trench structure along the fixed direction is 5% to 20% of a second projection area of the device unit along the fixed direction.

17. The semiconductor device of claim 11, wherein in a top view, a first projection area of the source trench structure along the fixed direction is 5% to 15% of a second projection area of the device unit along the fixed direction.

18. The semiconductor device of claim 11, wherein four sides of the hexagon have a first length, and the remaining two sides have a second length different from the first length.

19. The semiconductor device of claim 11, wherein the source trench structure comprises: a dielectric layer; and a conductor layer, surrounded by the dielectric layer.

20. The semiconductor device of claim 11, wherein the gate trench structure comprises: a gate dielectric layer; and a gate layer, surrounded by the gate dielectric layer.